AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MST224
Mathematical Methods

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 6/Dec. /2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **30**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1

[5×2 marks] Choose the correct answer:

1. The unit vector of $\mathbf{a} = 2\mathbf{b} - \mathbf{c}$, where $\mathbf{b} = \mathbf{i} + 2\mathbf{j} - \frac{5}{2}\mathbf{k}$, and $\mathbf{c} = -\mathbf{i} - 2\mathbf{j} - 3\mathbf{k}$, is:
   a) $\hat{\mathbf{a}} = \frac{1}{7}$
   b) $\hat{\mathbf{a}} = 7$
   c) $\hat{\mathbf{a}} = 1$
   d) $\hat{\mathbf{a}} = \frac{11}{7}$
   e) None of the above.

2. The volume of a parallelepiped with sides defined by vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j} + \mathbf{k}$, $\mathbf{b} = \mathbf{i} + \mathbf{j} - \mathbf{k}$ and $\mathbf{c} = 3\mathbf{i} - \mathbf{k}$, is:
   a) $V = 4$.
   b) $V = 5$.
   c) $V = 8$.
   d) $V = 0$.
   e) None of the above.

3. The particular integral of the differential equation $y'' - 2y' + y = (x-1)^2$, is:
   a) $y_p = (C + Dx)e^x$.
   b) $y_p = x^2 + 6x + 11$.
   c) $y_p = x + 1$.
   d) $y_p = x^2 - 2x + 1$.
   e) None of the above.

4. The angle between the two vectors $\mathbf{a} = 2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\mathbf{b} = 6\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$ in radian measure is:
   a) $\theta = -1.379$.
   b) $\theta = 2.342$.
   c) $\theta = -2.342$.
   d) $\theta = 1.379$.
   e) None of the above.

5. If $z = 2 - i$ then $|z|$ is:
   a) $|z| = 5$
   b) $|z| = \sqrt{5}$
   c) $|z| = 3$
   d) $|z| = \sqrt{3}$
   e) None of the above.

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PART 2

A) Find the general solution of the following differential equations:

i. **[4 marks]** $y'' + 2y' = 5\sin x$

ii. **[4 marks]** $xy' + y = 2x$.

iii. **[4 marks]** $y' = \frac{y\sin x}{1+2y^2}$.

B) [4 marks] Solve the initial value problem

$$\begin{cases} y' = y^2 + 4, \\ y(0) = -2. \end{cases}$$

C) [4 marks] Find the general solution of the system of differential equations

$$\begin{cases} \dot{x} = 3x + 2y, \\ \dot{y} = x + 4y. \end{cases}$$

End of questions

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